

Topological charge as a measurement principle for astrophysical $U(1)$ phase fields

A machine-checked chain – integrality, conservation by continuity, barrier, scale additivity, continuum limit – as the foundation of a measurement protocol, and a pre-registration template for its causal use

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SocrateAI-Scientific-QuantumFluids

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Abstract

Twenty-eight theorems, proved in Lean 4 against Mathlib and accepted by two independent kernels, form a chain about a $U(1)$ -valued phase read at the sample points of a closed loop: the loop sum of principal differences is an integer multiple of 2π ; it is conserved by every continuous evolution in which no sampled step reaches the branch point; a change of it forces such a step (a phase slip), which on an XY ring costs a positive energy barrier; the winding seen on the boundary of a region is the net charge of the plaquettes inside; and, for every fine enough sampling, the discrete winding equals the topological degree of the continuum phase along the loop. None of these statements mentions a superfluid. Their only structural input is $\pi_1(U(1)) = \mathbb{Z}$, so they hold for any $U(1)$ phase field: the phase of a complex scalar in a cosmic-string simulation, the axion field around a Peccei–Quinn string, the order parameter of a neutron-star superfluid. This report turns the chain into a measurement protocol with five principles, each carrying the exact hypothesis under which its theorem holds and the exact diagnostic that fires when the hypothesis fails; identifies three astrophysical settings where the hypotheses can be met; and supplies a pre-registration template for the one thing the chain does *not* give – the coupling of the integer to the observables – which must be measured, in each astrophysical model, by an energy-matched intervention with an equally arbitrary control arm. No astrophysical number is claimed here. The quantum-fluid work that produced the chain is used as the laboratory of the method, not as a source of values.

1 Scope and the rule that governs it

Everything below rests on a distinction that an earlier round of this project learned at some cost [1]: a mathematical structure shared by two systems is not a physical identification of the systems. The chain of theorems in Section 2 is shared by every $U(1)$ phase field because it is a theorem about $U(1)$; that is exactly why it transfers, and exactly why it says nothing, by itself, about what any particular field *does*. So:

Transfers. The proved chain, at the level of description where it is proved; and the method – a causal claim about a topological label needs a control arm that is equally arbitrary (the same surplus of the conserved quantities, in a different form).

Does not transfer. The coupling of the integer to the observables. It is empirical, it is being measured in one two-dimensional Bose gas with three base states [2], and it must be measured again inside each astrophysical model by that model’s own matched intervention.

2 The chain, as five measurement principles

Let a closed loop be sampled at points $0, \dots, n$ with phases $\theta_i \in \mathbb{R}$ read from the field ($\theta_n = \theta_0$), and $\text{pd}(a, b) \in (-\pi, \pi]$ the principal difference. The *discrete winding* is $w = \frac{1}{2\pi} \sum_{i < n} \text{pd}(\theta_i, \theta_{i+1})$. The Lean modules are those of [2, 3]; footprints are {propext, Classical.choice, Quot.sound} throughout, and every module carries negative controls.

Principle 1 (integrality; `VortexWinding.loop_sum_eq_mul`). $w \in \mathbb{Z}$ for any closed loop of real phases, whatever the field. Hypothesis: none. Use: a non-integer output is a bug in the reader, never a property of the field.

Principle 2 (conservation by continuity; `TopologicalProtection`). If the sampled phases depend continuously on the evolution parameter (time, conformal time, a sequence of simulation snapshots interpolated continuously) and no sampled step reaches π during the interval, then w is constant on the interval. Contrapositive: a change of w between two snapshots certifies that some step reached π in between – a phase slip on that loop. Hypothesis: continuity, and the no-slip condition. Diagnostic: a step whose forward and backward principal differences do not cancel (`pdiff_add_rev_eq_two_pi_iff`) is exactly a step at π ; it flags either a slip or an unresolved loop.

Principle 3 (barrier; `TopologicalProtection.mountain_pass`, `barrier_pos`). On a ring with nearest-neighbour cosine coupling J , any continuous evolution that changes w passes through a state of energy at least $-(n-2)J$; from the uniformly twisted $w = 1$ state the barrier is at least $2J - 2\pi^2 J/n > 0$ for $n \geq 10$, and the bound is false at $n = 9$. Hypothesis: the XY form of the gradient energy on the loop. Use: a lower bound on the energy an event must supply to change w ; events below it are not topological.

Principle 4 (scale additivity; `ScaleResolvedWinding.stokes`). For a region of plaquettes whose interior edges are paired by reversal, if no interior edge sits at π , the oriented boundary sum equals 2π times the net charge of the plaquettes inside. A neutral set of charges is invisible from every enclosing loop; a single charge is visible from every enclosing loop. Hypothesis: the no-cut condition on interior edges. Use: the net charge of a region is measurable from its boundary alone, and the scale-resolved quantity $W(R)$ (net charge inside R) is the variable that carries the physics, not the count of defects.

Principle 5 (continuum limit; `ContinuumWinding.discrete_eq_degree`, `detector_correct`). For a continuous closed loop of angles there is an integer w (the degree of a continuous lift) and a threshold n_0 such that every uniform sampling with $n \geq n_0$ points, with any choice of representatives of the sampled angles, returns exactly w . For a nowhere-vanishing closed loop of complex field values, the reader applied to `arg` returns the degree for all $n \geq n_0$. Hypothesis: the field does not vanish on the loop; the sampling is fine enough. Use: refinement is a test – if w changes under refinement the loop was unresolved; and a zero of the field on the loop is where the degree ceases to exist, which is what a slip is in the continuum.

2.1 What the chain does not contain

The argument principle (that the degree counts zeros inside the loop) is not proved; Principle 4 is its discrete substitute. The threshold n_0 is existential; a quantitative one needs a bound on $|\nabla\theta|$ away from cores. Principle 3 is proved for a global phase with XY energy; it is not proved for a *gauged* $U(1)$, where the physical winding is that of the covariant phase and needs the connection – see Section 6.

3 Three astrophysical settings where the hypotheses can be met

S1. Global and axion strings in lattice simulations. A complex scalar ϕ evolved on a grid after a Kibble–Zurek transition [4] forms a network of strings located, in practice, by

exactly the plaquette winding reader of Principle 1 [5, 6]; axion-string simulations [7] do the same for the Peccei–Quinn phase. Here every hypothesis of Principles 1, 2, 4 and 5 is met by construction: the field is a continuous function of conformal time, it is sampled on a grid, and its zeros are the string cores. What the chain adds to current practice is (i) a correctness theorem for the reader (Principle 5), with refinement as its test; (ii) the slip diagnostic of Principle 2, which certifies reconnection and loop-collapse events on a loop from two snapshots; (iii) boundary-only measurement of the string charge threading a region (Principle 4), which is what a flux-through-a-surface estimate needs; and (iv) the scale-resolved $W(R)$ as the natural variable for the scaling regime, in place of string length per volume.

S2. Neutron-star superfluids and glitches. The interior superfluid of a neutron star carries its rotation as an array of quantised vortices, $N_v = 2\Omega/\kappa$, and pulsar glitches are read as the catastrophic unpinning of many of them [8]. The persistent current of Principle 2 and the barrier of Principle 3 are the one-dimensional model of pinning: a vortex crossing a loop changes that loop’s w by one, and cannot do so without the energy of the slip. The chain gives the *measurement* side – what a glitch is on a loop, and what it costs – not the microphysics of pinning, which is where the observable numbers live and where a matched intervention would have to be designed inside a crust model.

S3. Superfluid and Bose–Einstein-condensate dark matter. If the halo is a condensate, its rotation is carried by vortices [9]. Principle 4 states what an observation of the halo’s outer velocity field can and cannot determine: the net charge inside, and nothing about the arrangement of bound pairs within. The measurement design here is the same as in S1, on the simulated halo field.

4 The protocol

1. **Read** w on every loop of interest by the principal-branch sum (Principle 1).
2. **Check resolution** by refinement: read again at twice the sampling; a change disqualifies the loop (Principle 5). Independently, flag every step whose forward and backward differences do not cancel (Principle 2’s diagnostic).
3. **Localise charge from boundaries:** tile the region, read plaquette charges, and verify that the boundary sum equals the interior total (Principle 4). A mismatch is an interior edge at π – an unresolved core – and is reported as such, not averaged away.
4. **Track** w across snapshots; every change is a certified slip on that loop (Principle 2), to be matched against an energy budget (Principle 3).
5. **Report** $W(R)$ as a function of scale, not defect counts.

Each step’s failure mode is a theorem’s hypothesis failing, and is reported by name.

5 The pre-registration template for the causal step

The chain establishes that w is a stable, scale-resolved, correctly readable label. Whether it is a *difference-maker* for an astrophysical observable is a separate, empirical question, and it is answered only by an intervention with a matched control. The template, to be filled and committed before any run of the astrophysical model:

1. **Base states.** Equilibrated (or scaling-regime) states of the model, saved.

2. **Arm V.** Impose a topological label – a string loop, a vortex array, a winding on a ring – by an exact construction whose correctness is checked by Principle 1 (the reader finds exactly what was imposed) and by a periodicity or boundary check, before any evolution.
3. **Arm P.** Add the *same* surplus of every conserved quantity the model has (energy, particle number, momentum, angular momentum), in a non-topological form, with the surplus matched by bisection to machine precision.
4. **Arm 0.** The base continued untouched.
5. **Predictions**, with thresholds fixed now, on the observables the astrophysical question cares about, over a window fixed now; and the pre-declared reading of any thermometer-like quantity that the P arm is expected to shift by construction.
6. **Effect size.** The effect per unit of surplus: the same surplus in V and P, compared on the observables. A large effect in V and a small one in P for the same surplus is the signature of a non-energetic difference-maker.
7. **Kill rule.** If V and P cannot be matched, or if Principle 1’s check fails on the imposed configuration, no causal claim is made.

The two-dimensional Bose gas of [2] is where this template was first executed; its numbers do not carry over, and its role here is to show that every step is executable.

6 Limits, stated before anyone asks

- **Gauged $U(1)$.** For local strings (Abelian–Higgs, Nielsen–Olesen) the physical winding is that of the gauge-covariant phase and involves the gauge connection. The chain is proved for a global phase. It applies to global and axion strings, and to a gauged theory only in a fixed gauge where the reader is applied to the gauge-fixed phase – which must be said.
- **Three dimensions.** The scale-additivity theorem is proved for plaquette regions (surfaces). Its use in three dimensions is on loops and on the surfaces they bound; that vortex lines do not end (`VortexWinding.balanced_sum_eq_zero`) is proved, the full three-dimensional flux accounting is not.
- **Continuity in time.** Principle 2 needs the sampled phases to be continuous in the evolution parameter. Simulations provide this; observations of a single epoch do not, and the chain then gives Principles 1, 4 and 5 only.
- **No numbers.** Nothing here predicts a string tension, a glitch size, or a halo rotation curve. Those are link 4, and link 4 is measured, not derived.

7 Literature status

The reader of Principle 1 is the standard string locator [5]; phase slips and persistent currents are Anderson’s [10]; the glitch model is Anderson and Itoh’s [8]; the string-network programme is reviewed in [6] and its axion version in [7]; the interventionist notion of cause is Woodward’s [11]. What is new here is the machine-checked status of each principle, the named diagnostic attached to each hypothesis, and the insistence that the causal step be pre-registered with a matched control inside the astrophysical model itself.

References

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